DATA ANALYSIS
SAC Revision – Part A   Name: _________________

Immunisation is a key aspect of public health policy for disease control around the world. In particular, data on immunisation rates is an important part of analysis. In this task we investigate the percentage of children aged 12-23 months immunised against the disease of measles. For this investigation use the data from Table 1.

How have the immunisation percentages changed for children between 1990 and 2014?

a. Complete a univariate analysis of the immunisation percentages for children for 1990 and 2014. Include in your analysis any graphs, diagrams or box plots that you decide would be of benefit to you in highlighting differences and similarities between the two years. Discuss why you decided to use your chosen graph(s) and displays for the data set.

b. Compare the two distributions in terms of shape, centre, spread and outliers. Discuss whether an association exists between the immunisation percentages for children over the two years.

Table 1: The percentage of children who received the measles vaccination in 2014 and 1990 in 30 countries.

Country Name	**Hemisphere**	**% of children immunised 1990**	**% of children immunised 2014**
Afghanistan	Northern	20	66
Argentina	Southern	93	95
Australia	Southern	86	93
Brazil	Southern	78	99
Cambodia	Northern	34	80
Canada	Northern	89	90
Colombia	Northern	82	91
Costa Rica	Northern	90	95
Denmark	Northern	84	90
Ethiopia	Northern	38	70
Fiji	Southern	84	94
Greece	Northern	76	97
Grenada	Northern	85	94
Guinea	Northern	35	52
Hungary	Northern	99	99
India	Northern	56	85
Ireland	Northern	78	93
Mali	Northern	43	80
Mauritius	Southern	76	98
Mexico	Northern	75	97
Morocco	Northern	79	99
Nepal	Northern	57	88
New Zealand	Southern	90	93
Samoa	Southern	89	82
Singapore	Northern	84	95
Switzerland	Northern	90	93
Thailand	Northern	80	99
United Kingdom	Northern	87	93
United States	Northern	90	92
Vietnam	Northern	88	97

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Your investigation may include the following:

Question 1.
Complete a univariate analysis of the immunisation percentages for children for 1990 and 2014.

a) Classify the data using categorical nominal, categorical ordinal, numerical discrete and numerical continuous.

*Country name:*____

Percentage of children immunised:____

b) Complete the frequency table, summarising the data collected for the *percentage of children in 1990*.

% of children	**Frequency**
0 –	
20 –	
40 -	
60 –	
80 – 100	
TOTAL:	

c) Using the frequency table from part **b**, prepare a graphical display showing the percentage of children immunised in 1990. (Hint: draw a histogram to display the percentage of children immunised in 1990.)

d) Use the histogram to describe the distribution, centre and spread of the data.

e) Would the mean or median be a better measure of central tendency for this data? Explain your answer. What is the value of the appropriate measure of centre (median or mean) for your 1990 data?

f) Is it appropriate to calculate z-scores for the percentage of children immunised in 1990 data? Explain your answer.

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Question 2
Complete a univariate analysis of the immunisation percentages for children for 1990 and 2014.
Repeat Question **1 parts b-f** for 2014.

Question 3

a) Calculate the five number summary for the percentage of children immunised in 1990 and in 2014.

	% of children immunised in 1990	**% of children immunised in 2014**
Min:		
Q1:		
Median:		
Q3:		
Max:		

b) Prepare a graphical display showing the percentage of children immunised in 1990 and in 2014. Discuss why you decided to use your chosen graphical display for the data set. **(HINT:** Using your CAS calculator prepare a display of parallel box plots showing the percentage of children immunised in 1990 and in 2014.)

c) Compare the two distributions in terms of shape, centre, spread and outliers.

d) Does there appear to be an association between the percentage of children immunised in 1990 and in 2014? Discuss whether an association exists between the immunisation percentages for children over the two years. Explain giving reference to an appropriate statistic.

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Question 4

a) Using the 30 countries complete the following two-way table using one year of data.
Data is for year 1990.

Percent immunised	**Hemisphere**	
	Northern	**Southern**
Low (0% -)		
Medium (35% -)		
High (70%-<100%)		
Total		

b) Use the data in the table above to complete the percentage two-way table.

Percent immunised	**Hemisphere**	
	Northern	**Southern**
Low (0% -)		
Medium (35% -)		
High (70%-<100%)		
Total	100	100

c) Prepare a graphical display for this two-way frequency table. Discuss why you decided to use your chosen graphical display for the data set. (HINT: Convert this data into a segmented bar chart).

d) Is there an association between the percentage of children immunised and the hemisphere in which they live? Write a brief response quoting appropriate statistics.

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